

Computer Organization & Assembly Language

Lab 11

Lab Instructor: Mr. Tariq Mehmood Butt

Teacher Assistants: Muhammad Umair bsdsf23m006@pu.edu.pk
Muhammad ZohaiB bsdsf23m023@pu.edu.pk

Topic: String Instructions

Task 1:

Write an assembly program that copies a `STRING_1` into another `STRING_2` in reverse order using **MOVSB** instruction.

Task 2:

Write an assembly program that takes a character string input from the user, stores it in a variable using the **STOSB** instruction, **allows corrections using the Backspace key**, stops input when **Enter** is pressed, and then displays the stored string using the **LODSB** instruction.

Task 3:

Write an assembly program that takes a target character as input. Use **SCASB** instruction to find the first occurrence of the target character in a string and displays the position in the string if the character is found, or display "Character is not found." otherwise.

Task 4:

Write an assembly program that takes a target character as input. Use the **SCASB** instruction to scan a character string and count the total number of times the target character appears in the string. Display the count.

Task 5:

Write an assembly program that compares two same length strings character by character using **CMPSB** instruction and displays the first mismatched character along with its position in the string.

Task 6:

Write an assembly program that scans a string using **SCASB** to find all vowels (A,E,I,O,U,a,e,i,o,u) and counts them. Display the vowels count.